perplexity

# Similarly, give me the complete list of locally-stored models I can use, and mention their hardware requirements and maximum concurrent user-handling capacity (go up to 800). No upper limit of hardware requirements.

| Model | RAM (Q4 Quant) | VRAM (Q4 Quant) | Concurrent Users (1x RTX 4090) | Concurrent Users (4x RTX 4090) | Concurrent Users (8x H100) | |
|---|---|---|---|---|---|---|
| Phi-4-mini | 4GB | 3GB | 50-100 | 200-400 | 400-800+ | |
| Llama 3.2 3B | 6GB | 4GB | 40-80 | 160-320 | 320-640+ | |
| Gemma 3 4B | 6GB | 4GB | 35-70 | 140-280 | 280-560+ | |
| Mistral 7B | 8GB | 5GB | 30-60 | 120-240 | 240-480+ | |
| Phi-4 14B | 12GB | 10GB | 25-50 | 100-200 | 200-400+ | |
| Llama 3.1 8B | 10GB | 8GB | 20-40 | 80-160 | 160-320+ | |
| Qwen 3 8B | 10GB | 8GB | 20-40 | 80-160 | 160-320+ | |
| Gemma 3 12B | 14GB | 12GB | 15-30 | 60-120 | 120-240+ | |
| Mistral Nemo 12B | 14GB | 12GB | 15-30 | 60-120 | 120-240+ | |
| Llama 3.1 70B | 40GB | 35GB | 5-10 | 20-40 | 40-80+ | |
| Qwen 3 32B | 28GB | 24GB | 8-15 | 32-60 | 64-120+ | |
| DeepSeek R1 32B | 28GB | 24GB | 8-15 | 32-60 | 64-120+ | |
| Llama 3.3 70B | 45GB | 40GB | 3-8 | 12-32 | 24-64+ | |
| Qwen 2.5 72B | 48GB | 42GB | 3-8 | 12-32 | 24-64+ | |
| GPT-OSS 120B | 65GB | 55GB | 1-3 | 4-12 | 8-24+ | [1] [2] [3] |

## How to use these models

**Pull command:** `ollama pull <model_name>` (e.g. `ollama pull llama3.1:8b`)

**Set in your code:** Change `OLLAMA_MODEL = "llama3.1:8b"` in `main.py`

## Key recommendations for your HR chatbot

| Scenario | Recommended Model | Why |
|---|---|---|
| Budget server (8-16GB RAM) | `phi-4-mini` or `llama3.2:3b` | Handles 50+ concurrent users, fast enough for chat |
| Production (24GB VRAM) | `llama3.1:8b` or `qwen3:8b` | Best quality/speed balance, 20-40 users per 4090 |
| High volume (64GB+ RAM) | `gemma3:12b` or `mistral-nemo:12b` | 15-30 users per 4090, excellent policy/reasoning |
| Enterprise (160GB+ RAM) | `llama3.1:70b` | Highest quality, handles complex policy inference |

## Concurrency scaling notes

- **Numbers assume Q4_K_M quantization** (good quality/speed balance)
- **1x RTX 4090** = baseline for most deployments
- **4x RTX 4090** = scales linearly with `OLLAMA_NUM_PARALLEL=4`
- **8x H100** = enterprise scale (set `OLLAMA_MAX_LOADED_MODELS=8`)
- **RAM overflow kills performance** — always stay within VRAM limits
- **Context window grows with users** — 4 users = 4x context usage [1] [4] [5]

All models above are **excellent for RAG/policy Q&A.** `llama3.1:8b` is the sweet spot for most HR chatbot deployments.[3] [6] [1]

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1. https://localllm.in/blog/ollama-vram-requirements-for-local-llms
2. https://www.reddit.com/r/LocalLLaMA/comments/1qu3ldm/rag_chat_with_your_documents_34_concurrent_users/
3. https://localaimaster.com/blog/small-language-models-guide-2026
4. https://docs.ollama.com/faq
5. https://ollama.readthedocs.io/en/faq/
6. https://yourgpt.ai/blog/growth/best-llms
7. https://github.com/infiniflow/ragflow/issues/7981
8. https://www.reddit.com/r/ollama/comments/1qm9cgp/ollama_models_ranked_by_vram_requirements/
9. https://www.glukhov.org/post/2025/05/how-ollama-handles-parallel-requests/
10. https://ollama.com/blog/new-model-scheduling

11. https://iproyal.com/blog/best-local-llms/
12. https://www.mintlify.com/ollama/ollama/models
13. https://daily.dev/blog/running-llms-locally-ollama-llama-cpp-self-hosted-ai-developers
14. https://www.ikangai.com/the-complete-guide-to-running-llms-locally-hardware-software-and-performance-essentials/
15. https://www.youtube.com/watch?v=bzpRIF2Q16c